

Lab 1

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure for 'PostgreSQL 16', including 'dvdrental' and 'postgres' databases. The 'dvdrental' database is expanded, showing various objects like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, and Schemas. The 'public' schema is expanded, showing objects like Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, Tables, Trigger Functions, Types, and Views. The 'postgres' database is also expanded, showing objects like Casts, Catalogs, Event Triggers, and Extensions.

The main pane shows the 'Query' tab with the following SQL query:

```
1 CREATE ROLE cashier_role ;
```

The 'Data Output' tab shows the query result:

```
CREATE ROLE
```

The 'Messages' tab shows the query execution status:

```
Query returned successfully in 46 msec.
```

The status bar at the bottom indicates: 'Total rows: Query complete 00:00:00.046' and 'LF Ln 1, Col 27'.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure for 'PostgreSQL 16', including 'dvdrental' and 'postgres' databases. The 'dvdrental' database is expanded, showing various objects like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, and Schemas. The 'public' schema is expanded, showing objects like Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, Tables, Trigger Functions, Types, and Views. The 'postgres' database is also expanded, showing objects like Casts, Catalogs, Event Triggers, and Extensions.

The main pane shows the 'Query' tab with the following SQL queries:

```
1 CREATE ROLE cashier_role ;
2
3 CREATE USER mike WITH PASSWORD 'mike123';
4 CREATE USER jon WITH PASSWORD 'jon123';
```

The 'Data Output' tab shows the query result:

```
CREATE ROLE
```

The 'Messages' tab shows the query execution status:

```
Query returned successfully in 101 msec.
```

The status bar at the bottom indicates: 'Total rows: Query complete 00:00:00.101' and 'LF Ln 3, Col 1'.

Object Explorer

- Servers (1)
 - PostgreSQL 16
 - Databases (2)
 - dvrental
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions

Dashboard X dvrental/postgre... X dvrental/postgres@PostgreSQL 16* X

dvrental/postgres@PostgreSQL 16

Query Query History

```
1 CREATE ROLE cashier_role ;
2
3 CREATE USER mike WITH PASSWORD 'mike123';
4 CREATE USER jon WITH PASSWORD 'jon123';
5
6 GRANT cashier_role TO mike;
7 GRANT cashier_role TO jon;
```

Scratch Pad X

Data Output Messages Notifications

GRANT ROLE

Query returned successfully in 44 msec.

Total rows: Query complete 00:00:00.044

Query returned successfully in 44 msec. X

LF Ln 6, Col 1

Object Explorer

- Servers (1)
 - PostgreSQL 16
 - Databases (2)
 - dvrental
 - Castes
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 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - postgres
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions

Dashboard X dvrental/postgre... X dvrental/postgres@PostgreSQL 16* X

dvrental/postgres@PostgreSQL 16

Query Query History

```
1 CREATE ROLE cashier_role ;
2
3 CREATE USER mike WITH PASSWORD 'mike123';
4 CREATE USER jon WITH PASSWORD 'jon123';
5
6 GRANT cashier_role TO mike;
7 GRANT cashier_role TO jon;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT, INSERT ON payment TO cashier_role;
```

Scratch Pad X

Data Output Messages Notifications

GRANT

Query returned successfully in 45 msec.

Total rows: Query complete 00:00:00.045

Query returned successfully in 45 msec. X

LF Ln 9, Col 1

Object Explorer

- Servers (1)
 - PostgreSQL 16
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 - dvdrental
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 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
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 - FTS Configurations
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 - Trigger Functions
 - Types
 - Views

Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History

```
1 CREATE ROLE cashier_role ;
2
3 CREATE USER mike WITH PASSWORD 'mike123';
4 CREATE USER jon WITH PASSWORD 'jon123';
5
6 GRANT cashier_role TO mike;
7 GRANT cashier_role TO jon;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT, INSERT ON payment TO cashier_role;
11
12 SET ROLE mike;
13 SELECT COUNT(*) FROM payment; -- 14596 (thấy hết, cả của Jon)
14 RESET ROLE;
```

Data Output Messages Notifications

RESET

Query returned successfully in 45 msec.

Total rows: Query complete 00:00:00.045

✓ Query returned successfully in 45 msec. X

LF Ln 12, Col 1

Object Explorer

- Servers (1)
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 - dvdrental
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 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
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 - Domains
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Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History

```
1 CREATE ROLE cashier_role ;
2
3 CREATE USER mike WITH PASSWORD 'mike123';
4 CREATE USER jon WITH PASSWORD 'jon123';
5
6 GRANT cashier_role TO mike;
7 GRANT cashier_role TO jon;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT, INSERT ON payment TO cashier_role;
11
12 SET ROLE mike;
13 SELECT COUNT(*) FROM payment; -- 14596 (thấy hết, cả của Jon)
14 RESET ROLE;
15
16 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 95 msec.

Total rows: Query complete 00:00:00.095

✓ Query returned successfully in 95 msec. X

LF Ln 16, Col 1

Object Explorer

- Servers (1)
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 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
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 - Trigger Functions
 - Types
 - Views

Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History

```
6 GRANT cashier_role TO mike;
7 GRANT cashier_role TO jon;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT, INSERT ON payment TO cashier_role;
11
12 SET ROLE mike;
13 SELECT COUNT(*) FROM payment; -- 14596 (thấy hết, cả của Jon)
14 RESET ROLE;
15
16 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
17
18 CREATE POLICY isolate_cashier_payments ON payment
19 FOR SELECT
20 TO cashier_role
21 USING (
22 staff_id = (SELECT staff_id FROM staff WHERE lower(username) = current_user)
23 );
```

Data Output Messages Notifications

CREATE POLICY

Query returned successfully in 68 msec.

Total rows: Query complete 00:00:00.068

✓ Query returned successfully in 68 msec. X

Object Explorer

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 - Languages
 - Publications
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 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
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 - Types
 - Views

Dashboard X dvdrental/postgre... X dvdrental/postgres@PostgreSQL 16* X

dvdrental/postgres@PostgreSQL 16

Query Query History

```
16 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
17
18 CREATE POLICY isolate_cashier_payments ON payment
19 FOR SELECT
20 TO cashier_role
21 USING (
22 staff_id = (SELECT staff_id FROM staff WHERE lower(username) = current_user)
23 );
24
25 SET ROLE mike;
26 SELECT COUNT(*) FROM payment;
27 SELECT COUNT(*) FROM payment WHERE staff_id = 2;
28
29 SET ROLE jon;
30 SELECT COUNT(*) FROM payment;
31
32 RESET ROLE;
33 SELECT COUNT(*) FROM payment;
```

Data Output Messages Notifications

count	bigint
1	14596

Showing rows: 1 to 1 Page No: 1 of 1

Total rows: 1 Query complete 00:00:00.113

✓ Successfully run. Total query runtime: 113 msec. 1 rows affected. X